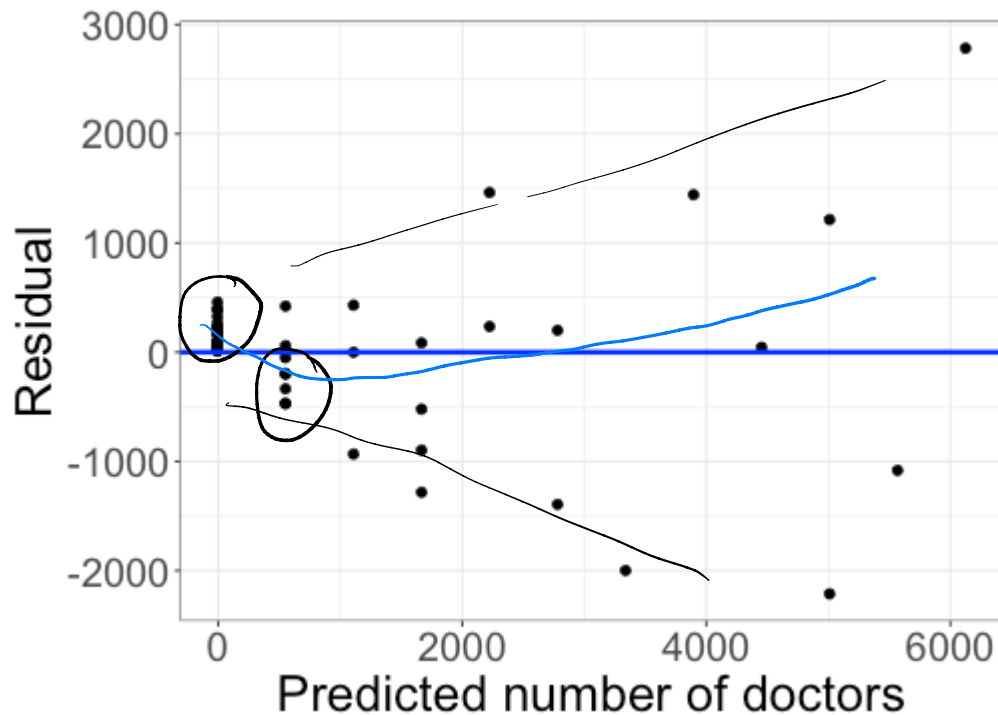


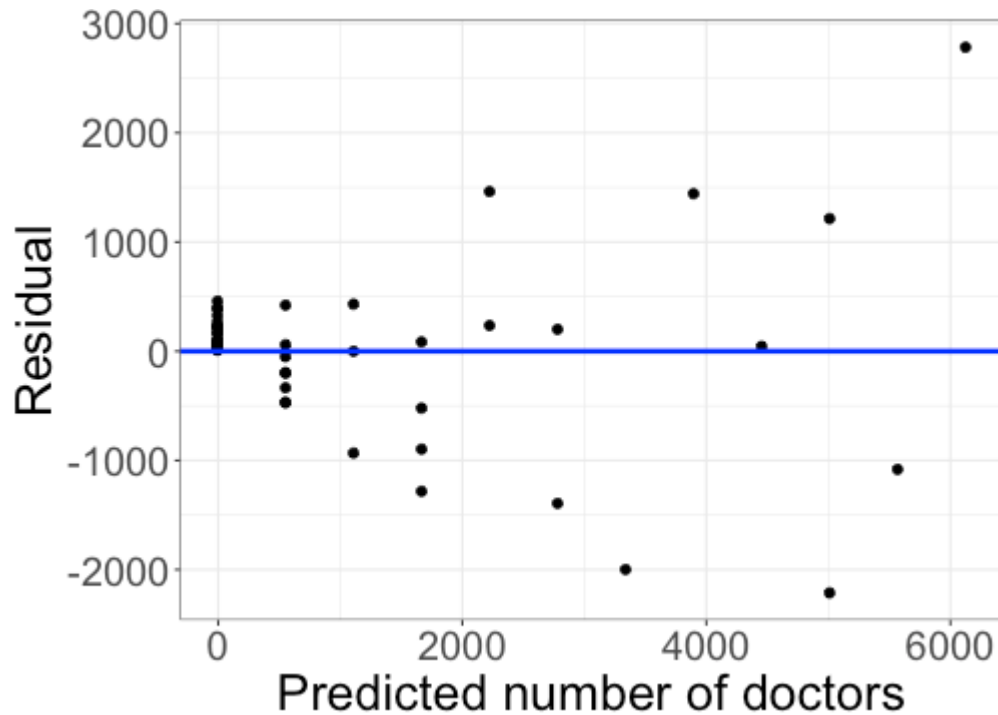
Nonlinear relationships and transformations

Last time



Do the shape and constant variance assumptions look reasonable?

Today

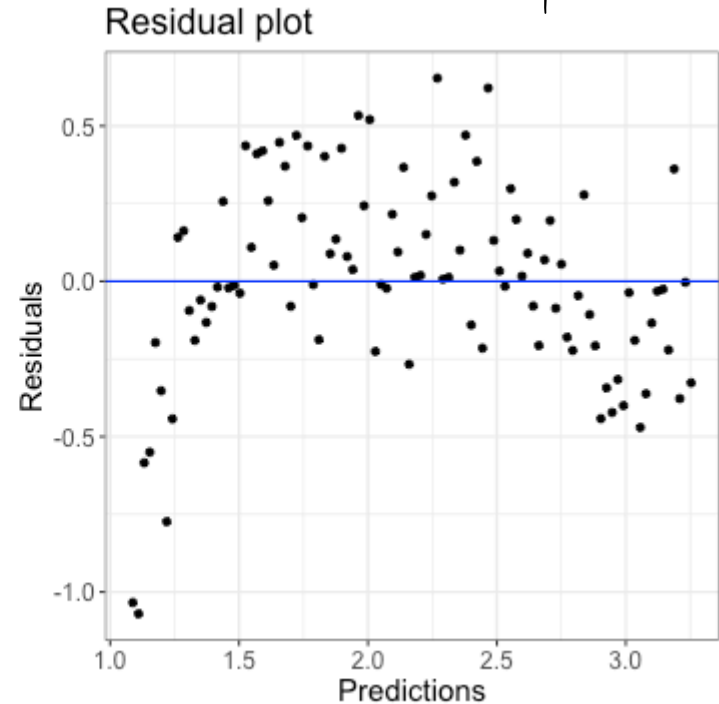
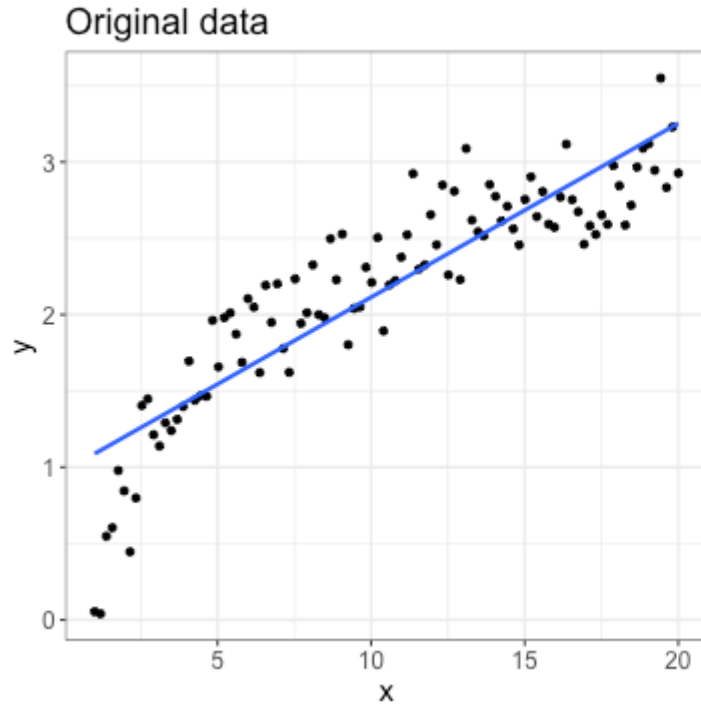


Today: handling non-linear relationships

Example

$$-x^2$$
$$\sqrt{x}$$

$$e^x$$
$$\log(x)$$



The relationship between x and y does not appear linear. How could we model that relationship instead?

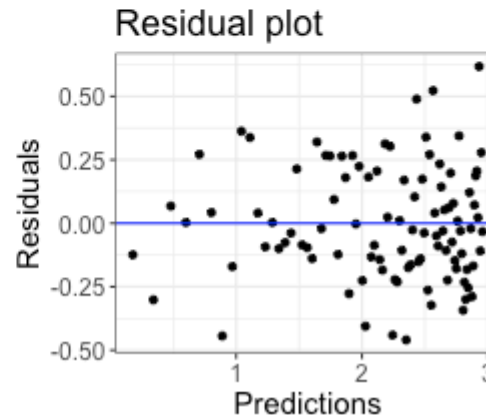
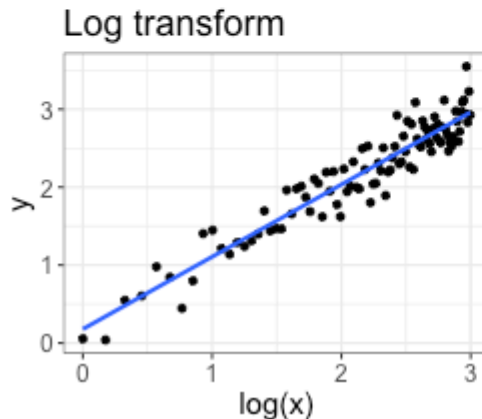
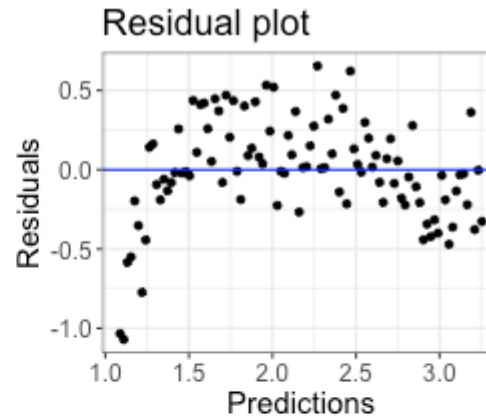
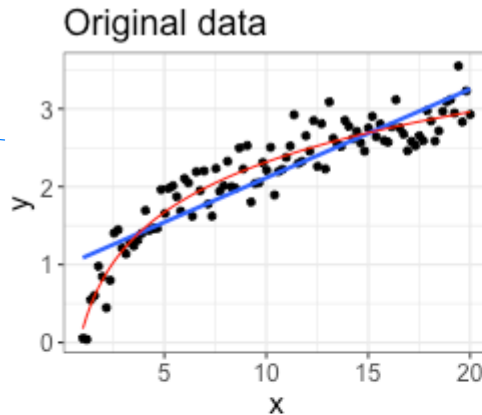
$$a = \log(b) \Leftrightarrow b = e^a$$

Example: log transform

$$Y = \beta_0 + \beta_1 \log(X) + \varepsilon$$

$\log_{10}$ $\log_e$ (natural log)
 $\uparrow$ aka $\ln$
 (base e)
 $\leftarrow$ not linear in X ,
 but is linear in $\log(X)$

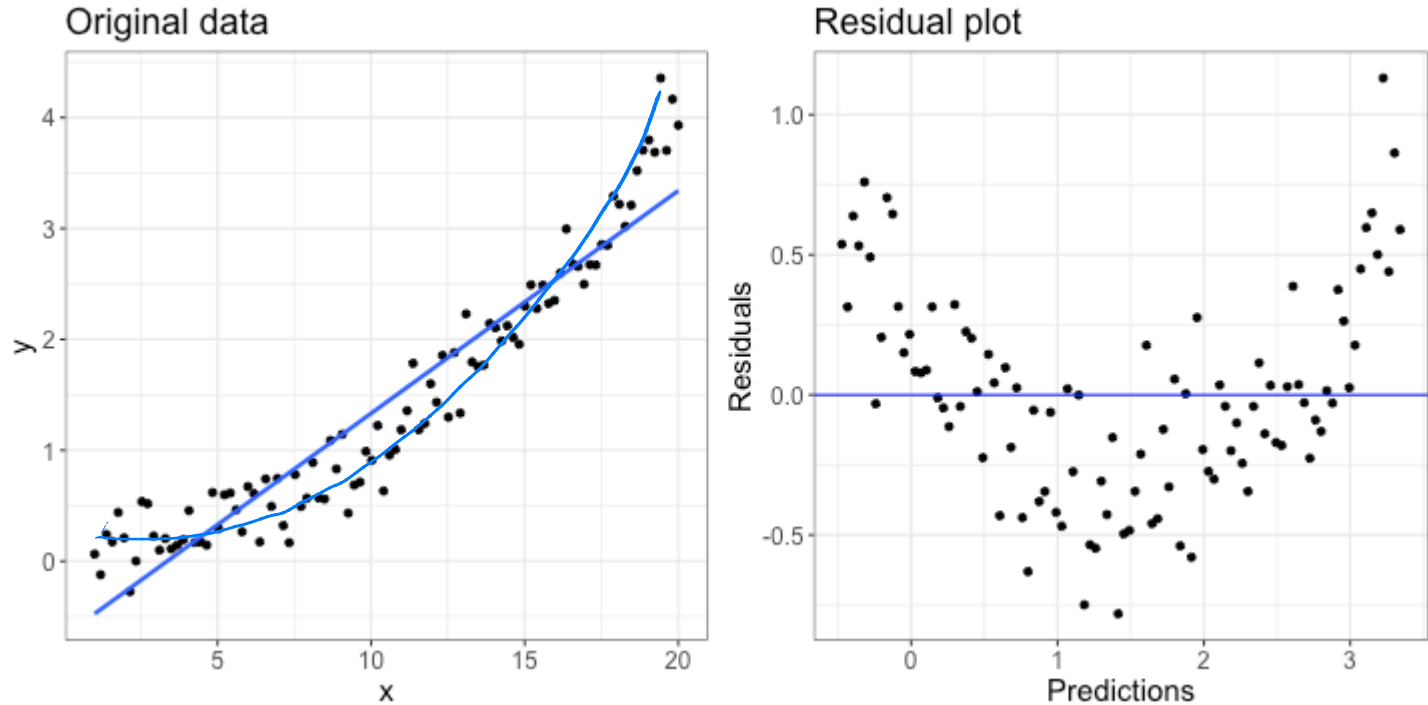
red: Fit w/ log
 blue: linear fit



looks better!

$$x^2$$

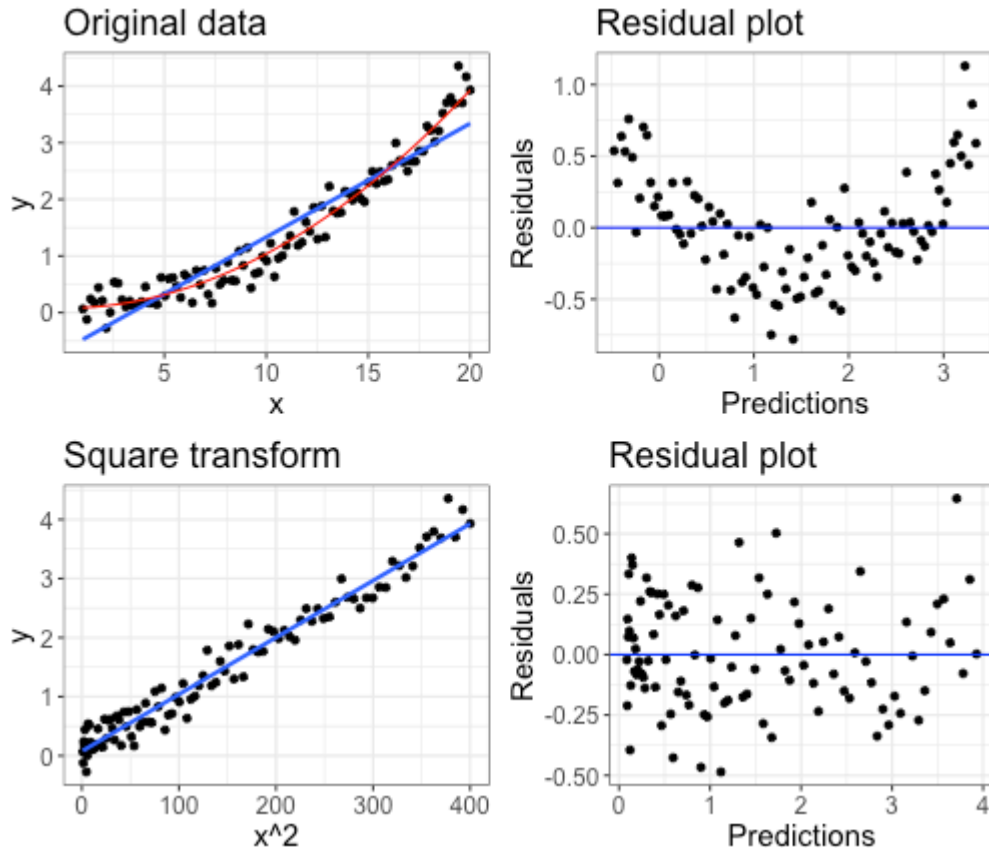
Example



The relationship between x and y does not appear linear. How could we model that relationship instead?

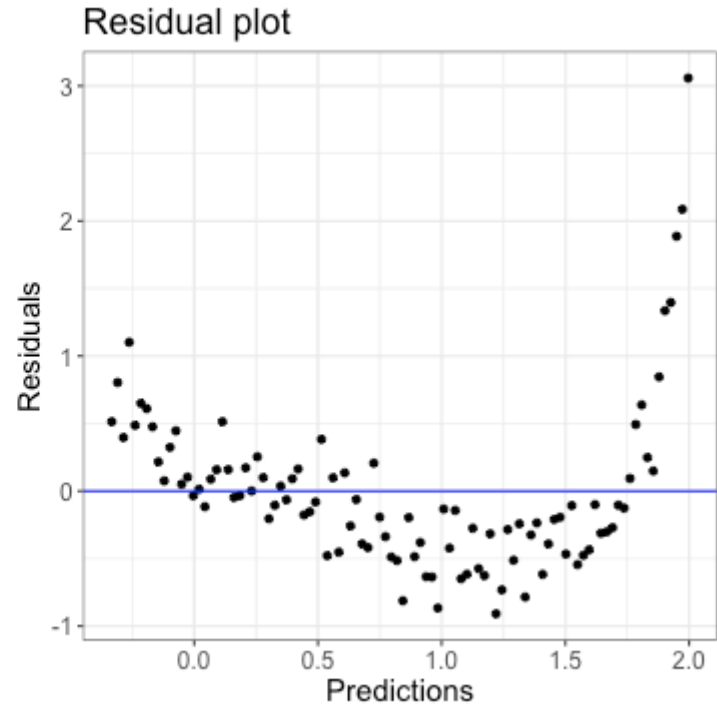
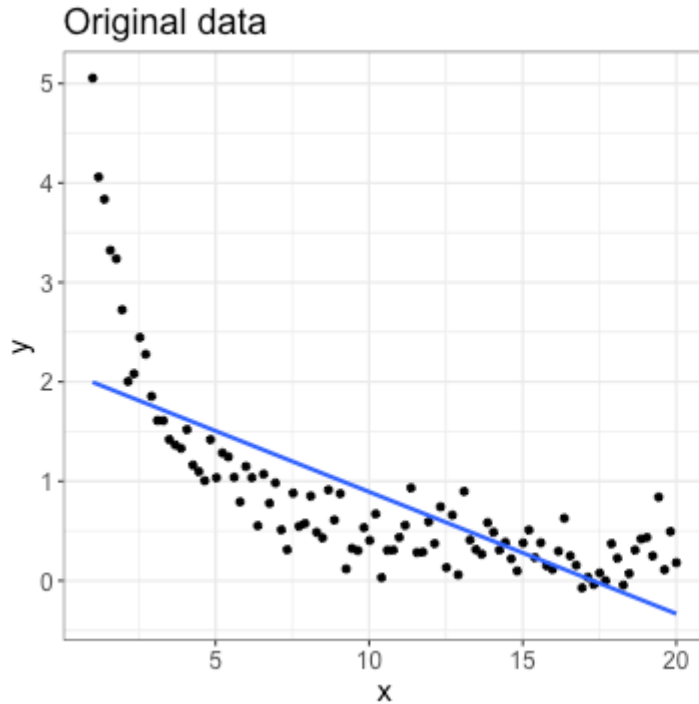
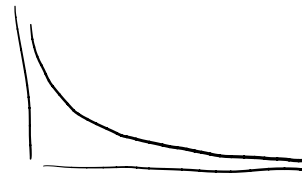
Example: squaring X

$$Y = \beta_0 + \beta_1 X^2 + \varepsilon$$



Example

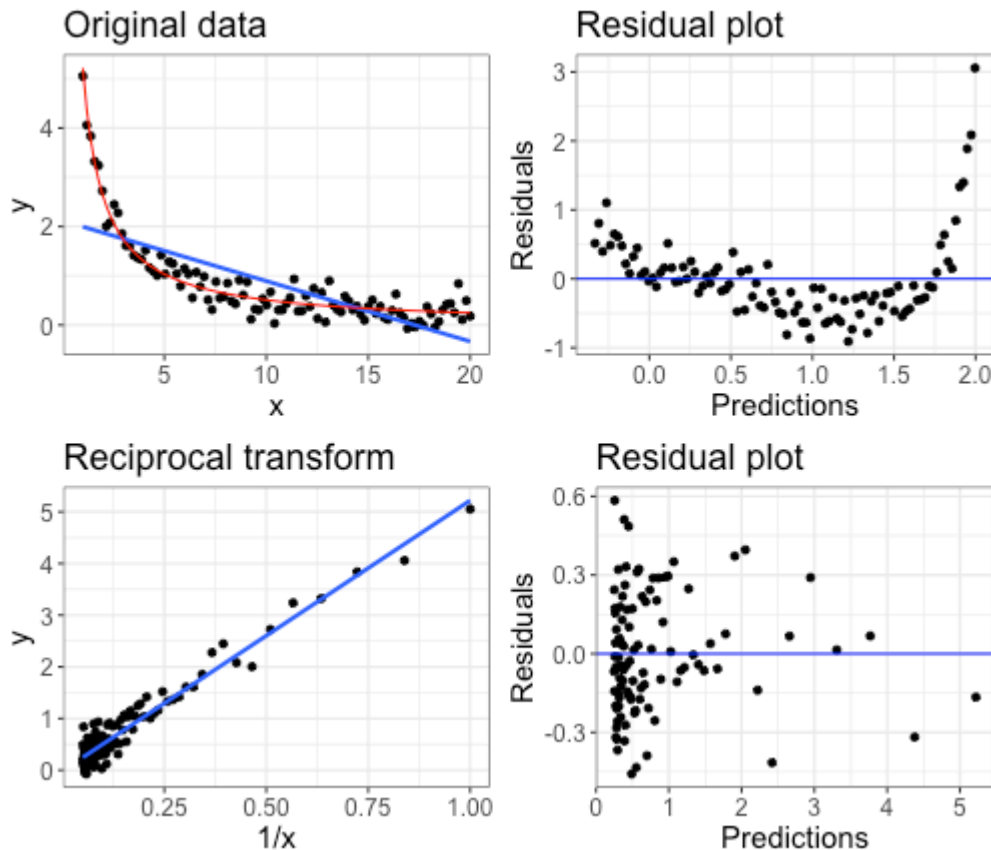
$\frac{1}{x}$



The relationship between x and y does not appear linear. How could we model that relationship instead?

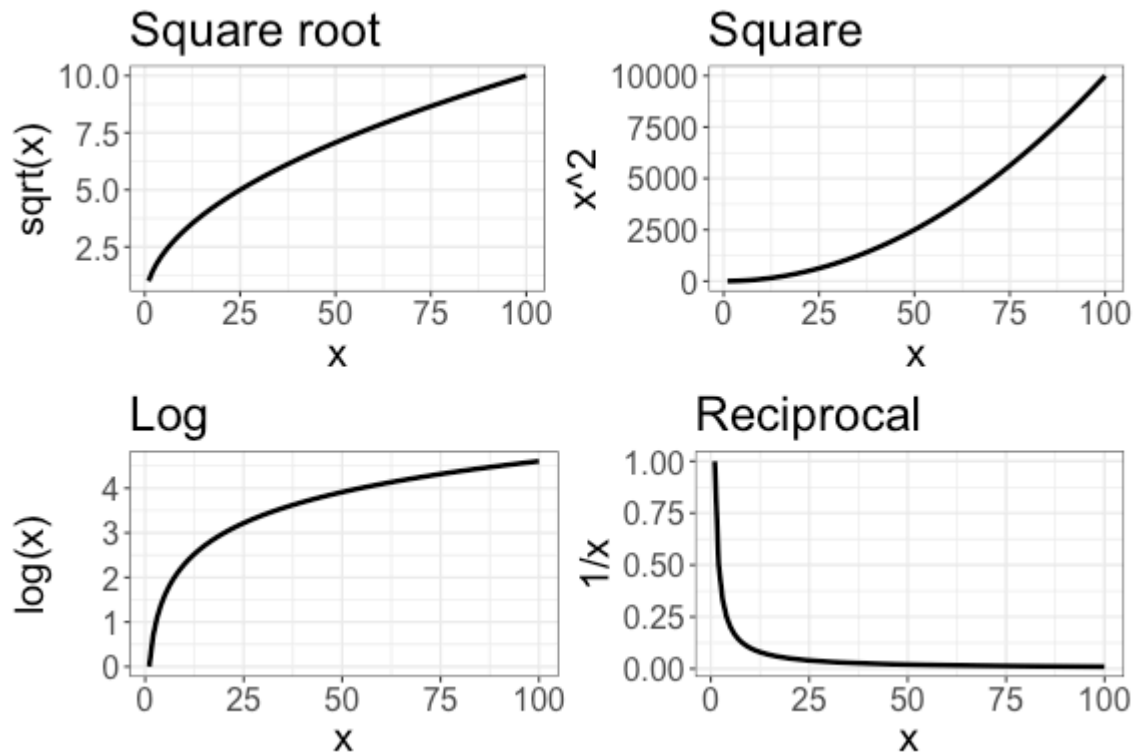
Example: reciprocal transform

$$Y = \beta_0 + \beta_1 (1/X) + \varepsilon$$

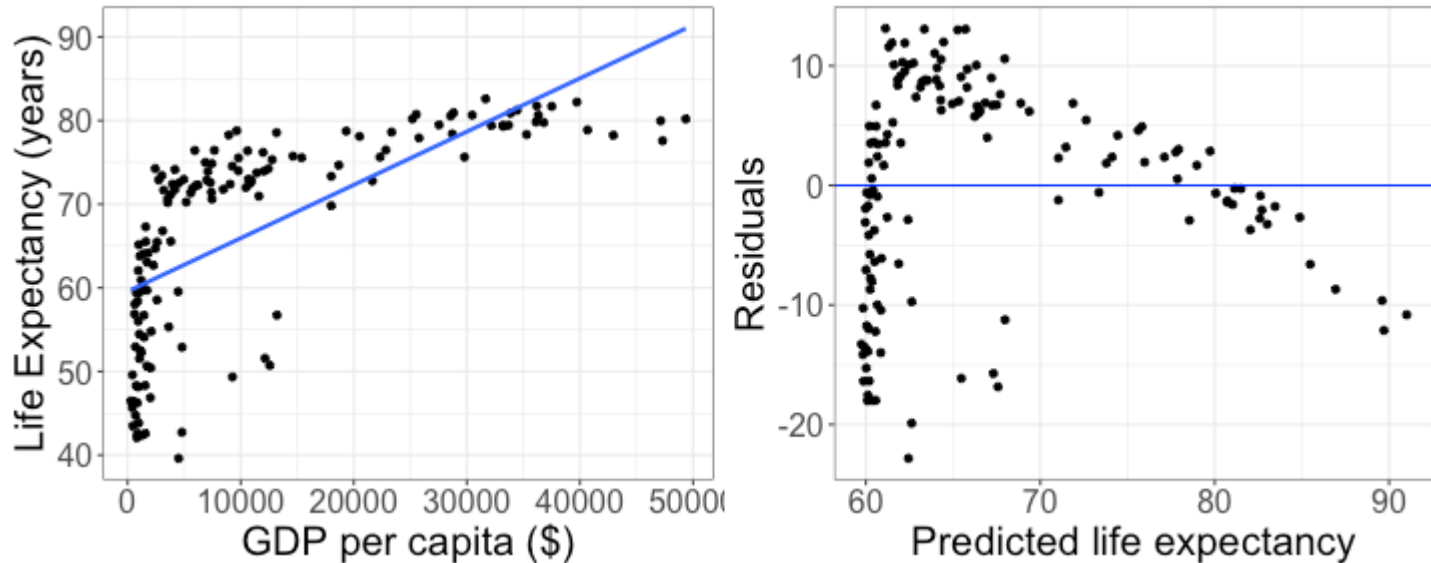


Choosing transformations

To model nonlinear relationships, try transformations for x that looks like the relationship

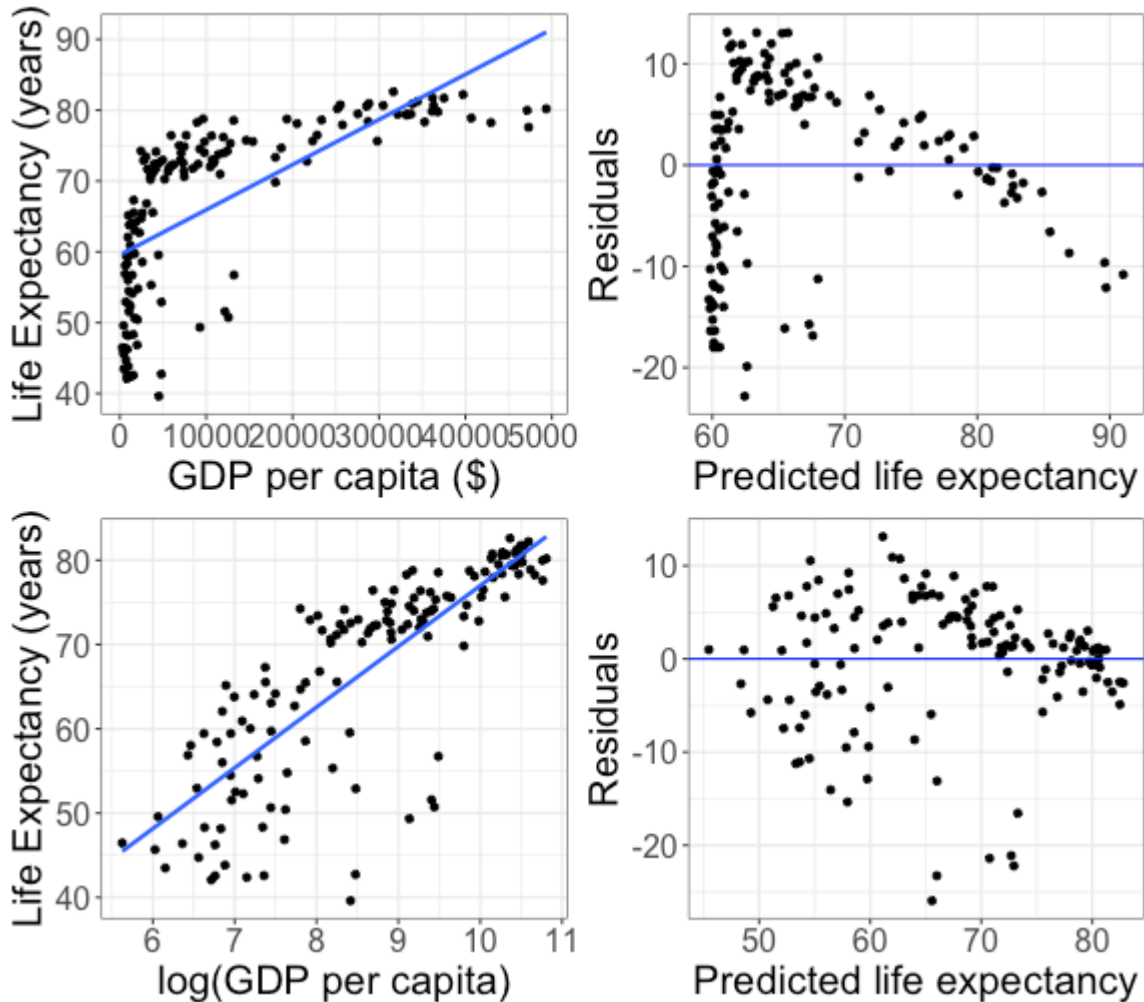


Example

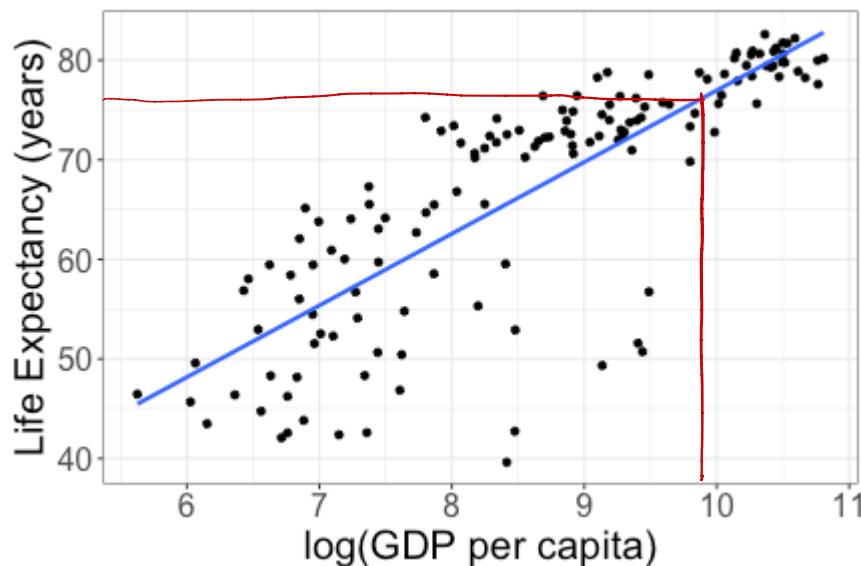


What transformation could I try for this data?

Log transformation



Log transformation



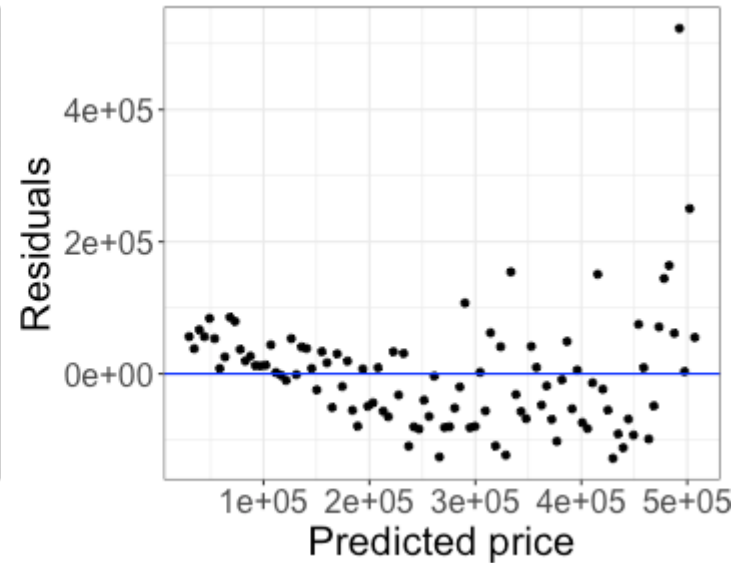
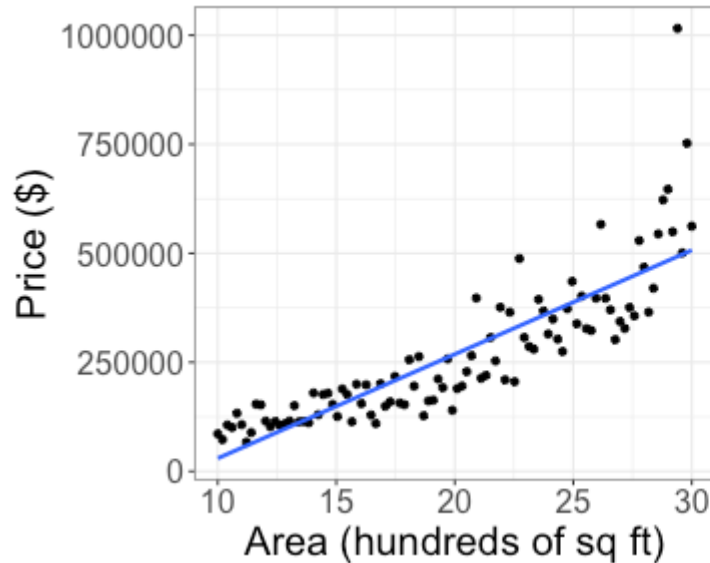
$$\widehat{\text{Life Expectancy}} = 4.95 + 7.20 \log(\text{GDP per capita})$$

What life expectancy would I predict for a GDP per capita of \$20000?

$$\begin{aligned} & 4.95 + 7.20 \log(20000) \\ &= 4.95 + 7.2(9.9) = 76.255 \text{ (years)} \end{aligned}$$

χ^2 e^x

Example

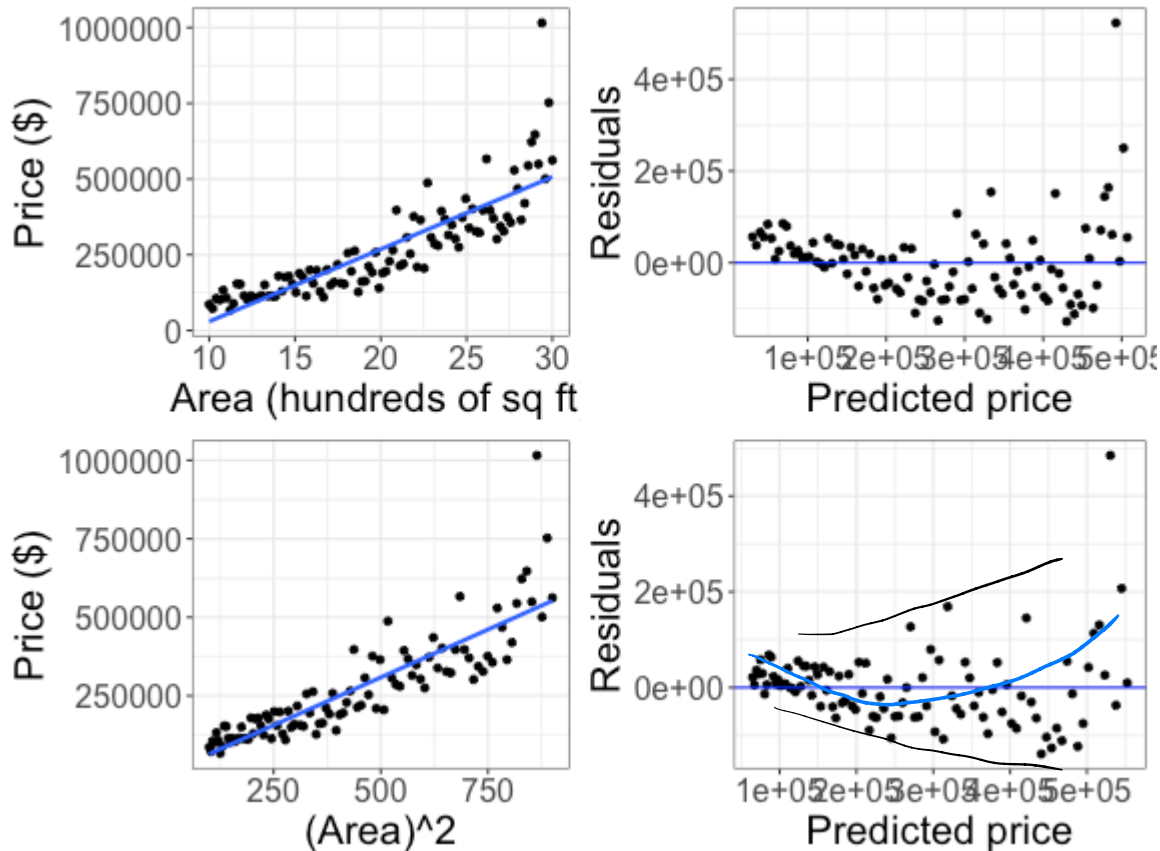


What transformation could I try for this data?

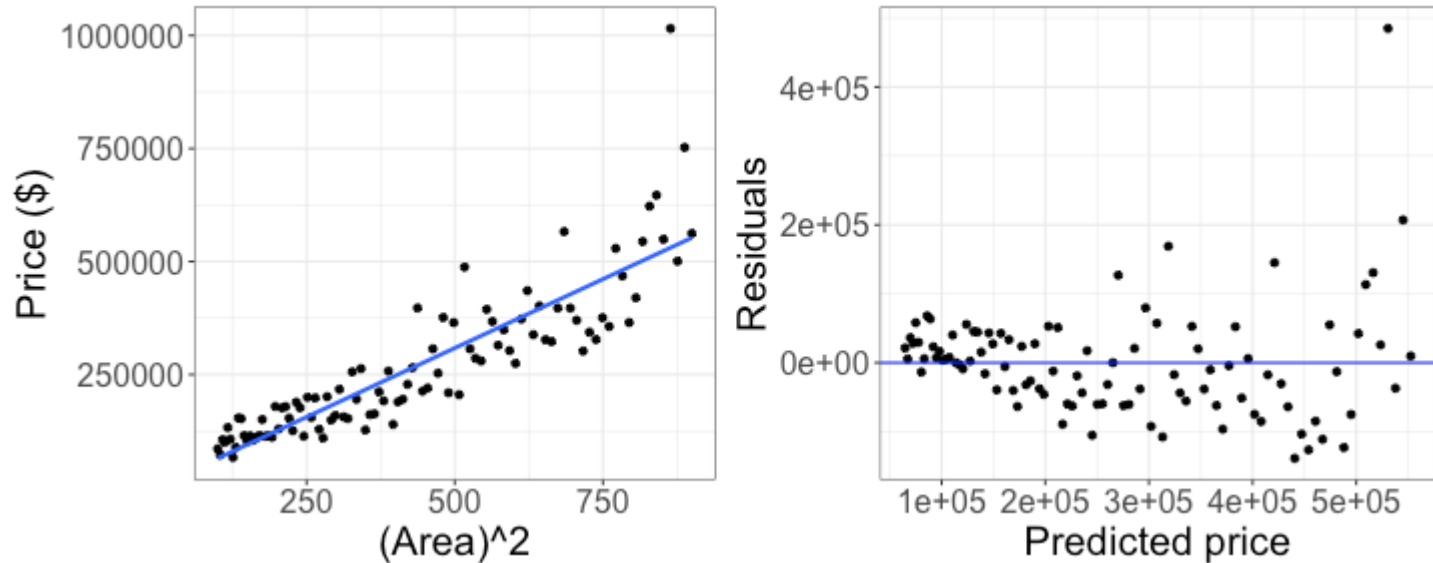
Trying squaring X

non-constant variance: might try
transforming y

$$\text{Model: } Price = \beta_0 + \beta_1 (Area)^2 + \varepsilon$$



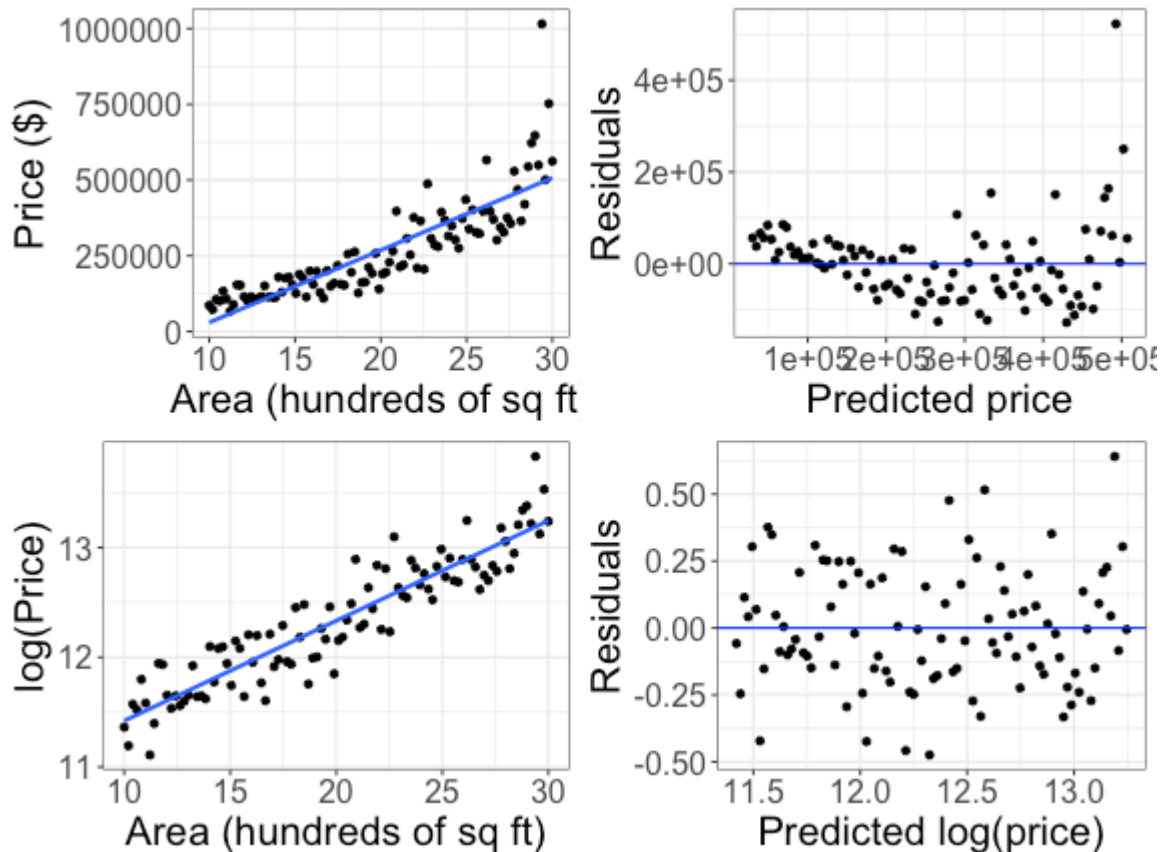
Trying squaring X

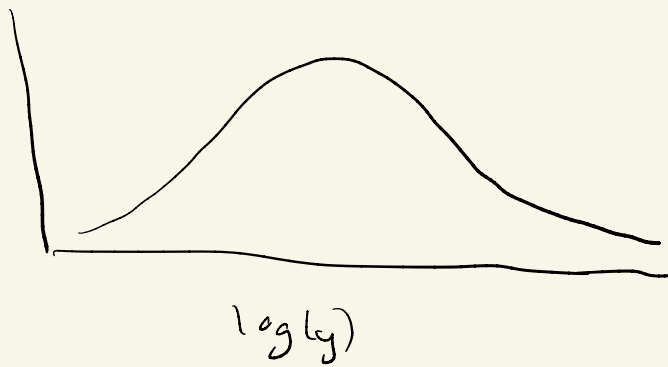


What do you notice about the residual plot after trying the transformation?

Another attempt: transforming the response!

$$\text{Model: } \log(\text{Price}) = \beta_0 + \beta_1 \text{Area} + \varepsilon$$





$$a = \log(b)$$

$$e^a = b$$

Log transformed response

$$\widehat{\log(\text{Price})} = 10.51 + 0.091 \text{ Area}$$

+ Here Price is in \$, Area is in hundreds of square feet

How would I calculate the predicted *price* (in \$) for a house of 2000 square feet?

↳ Area = 20

$$\widehat{\log(\text{Price})} = 10.51 + 0.091 (20)$$

$$= 12.33$$

$$\widehat{\text{Price}} = e^{12.33} = \$ 226\,386$$

Class activity

Work on the handout, then we will discuss as a class

Class activity

$$\widehat{\text{Life Expectancy}} = 4.95 + 7.20 \log(\text{GDP per capita})$$

What is the predicted price for GDP per capita of \$20200?

Class activity

$$\widehat{\text{LifeExpectancy}} = 4.95 + 7.20 \log(\text{GDP per capita})$$

- + When GDP per capita = 20000: $\widehat{\text{LifeExpectancy}} = 76.255$
- + When GDP per capita = 20200: $\widehat{\text{LifeExpectancy}} = 76.327$

Change:

Class activity

$$\widehat{\log(\text{Price})} = 10.51 + 0.091 \text{ Area}$$

+ Price in \$, Area in hundres of sq ft

What is the predicted price for a house of 2100 square feet?

Class activity

$$\widehat{\log(\text{Price})} = 10.51 + 0.091 \text{ Area}$$

- + Price in \$, Area in hundres of sq ft
- + When Area = 20: $\widehat{\text{Price}} = 226386$
- + When Area = 21: $\widehat{\text{Price}} = 247954$

Change:

Class activity

$$\log(\widehat{\text{Price}}) = 10.51 + 0.091 \text{ Area}$$

- + Price in \$, Area in hundreds of sq ft
- + When Area = 20: $\widehat{\text{Price}} = 226386$
- + When Area = 21: $\widehat{\text{Price}} = 247954$

$$\text{Change: } \frac{247954}{226386} = 1.095 = e^{0.091}$$